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        "from sklearn.model_selection import train_test_split\n",
        "from sklearn import svm\n",
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"2      8   183      64       0    0 23.3 \n",
"3      1    89      66      23   94 28.1 \n",
"4      0   137      40      35   168 43.1 \n",
"\n",
" DiabetesPedigreeFunction Age Outcome \n",
"0          0.627 50    1 \n",
"1          0.351 31    0 \n",
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"     <th>Glucose</th>\n",
"     <th>BloodPressure</th>\n",
"     <th>SkinThickness</th>\n",
"     <th>Insulin</th>\n",
"     <th>BMI</th>\n",
"     <th>DiabetesPedigreeFunction</th>\n",
"     <th>Age</th>\n",
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" fill: #FFFFFF;\n",
" }\n",
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" const buttonEl =\n",
" document.querySelector('#df-61fe7c86-71d2-4f22-bbfa-49b9e4a91
42f button.colab-df-convert');\n",
" buttonEl.style.display =\n",
" google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"\n",
" async function convertToInteractive(key) {\n",
" const element = document.querySelector('#df-61fe7c86-71d2-4f22-b
bfa-49b9e4a9142f');\n",
" const dataTable =\n",
" await google.colab.kernel.invokeFunction('convertToInteractive',\n",
" [key], {});\n",

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"    if (!dataTable) return;\n",
"\n",
"    const docLinkHtml = 'Like what you see? Visit the ' +\n",
"    '<a target=\"_blank\" href=https://colab.research.google.com/notebooks/data_table.ipynb>data table notebook</a>'\n",
"    + ' to learn more about interactive tables.';\n",
"    element.innerHTML = ";\n",
"    dataTable['output_type'] = 'display_data';\n",
"    await google.colab.output.renderOutput(dataTable, element);\n",
"    const docLink = document.createElement('div');\n",
"    docLink.innerHTML = docLinkHtml;\n",
"    element.appendChild(docLink);\n",
"  }\n",
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\n    \"max\": 99,\n    \"num_unique_values\": 51,\n    \"samples\":[\n      7,\n      12,\n      48\n    ],\n    \"semantic_type\":"\",\n    \"description\":"\"\n }
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"min     0.000000   0.000000   0.000000   0.000000   0.000000 \n",
"25%     1.000000   99.000000   62.000000   0.000000   0.000000 \n",
",
"50%     3.000000  117.000000   72.000000   23.000000  30.500000 \n",
"75%     6.000000  140.250000   80.000000   32.000000  127.250000 \n",
"max    17.000000  199.000000  122.000000   99.000000  846.000000 \n",
"\n",
"    BMI  DiabetesPedigreeFunction  Age  Outcome \n",
"count  768.000000           768.000000  768.000000  768.000000 \n",
"mean   31.992578           0.471876  33.240885   0.348958 \n",
"std    7.884160           0.331329  11.760232   0.476951 \n",
"min     0.000000           0.078000  21.000000   0.000000 \n",

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  " <th>BMI</th>\n",

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```

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```
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```
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60v-160H180v160Zm440 0h160v-160H620v160ZM180-180h160v-160H180v16
0Zm440 0h160v-160H620v160Z\"/>\n",
"    </svg>\n",
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"   background-color: #434B5C;\n",  
"   box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);\n",  
"   filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));\n",  
"   fill: #FFFFFF;\n",  
" }\n",  
" </style>\n",  
"\n",  
" <script>\n",
```

```

"    const buttonEl =\n",
"    document.querySelector('#df-fbf1531b-02eb-46da-a8e1-eee110af6
3c7 button.colab-df-convert');\n",
"    buttonEl.style.display =\n",
"    google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"\n",
"    async function convertToInteractive(key) {\n",
"    const element = document.querySelector('#df-fbf1531b-02eb-46da-
a8e1-eee110af63c7');\n",
"    const dataTable =\n",
"    await google.colab.kernel.invokeFunction('convertToInteractive',\n",
"    [key], {});\n",
"    if (!dataTable) return;\n",
"\n",
"    const docLinkHtml = 'Like what you see? Visit the ' +\n",
"    '<a target=\"_blank\" href=https://colab.research.google.com/noteb
ooks/data_table.ipynb>data table notebook</a>'\n",
"    + ' to learn more about interactive tables.';\n",
"    element.innerHTML = ";\n",
"    dataTable['output_type'] = 'display_data';\n",
"    await google.colab.output.renderOutput(dataTable, element);\n",
"    const docLink = document.createElement('div');\n",
"    docLink.innerHTML = docLinkHtml;\n",
"    element.appendChild(docLink);\n",
"    }\n",
"    </script>\n",
"    </div>\n",
"\n",
"\n",
"    </div>\n",
"    </div>\n"
],
"application/vnd.google.colaboratory.intrinsic+json": {

```

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"type": "dataframe",
"summary": "{\n  \"name\": \"diabetes_dataset\",\n  \"rows\": 8,\n  \"fields\": [\n    {\n      \"column\": \"Pregnancies\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 269.85223453356366,\n        \"min\": 0.0,\n        \"max\": 768.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          3.8450520833333335,\n          3.0,\n          768.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Glucose\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 243.73802348295857,\n        \"min\": 0.0,\n        \"max\": 768.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          120.89453125,\n          117.0,\n          768.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"BloodPressure\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 252.85250535810619,\n        \"min\": 0.0,\n        \"max\": 768.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          69.10546875,\n          72.0,\n          768.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"SkinThickness\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 263.7684730531098,\n        \"min\": 0.0,\n        \"max\": 768.0,\n        \"num_unique_values\": 7,\n        \"samples\": [\n          768.0,\n          20.536458333333332,\n          32.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Insulin\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 350.26059167945886,\n        \"min\": 0.0,\n        \"max\": 846.0,\n        \"num_unique_values\": 7,\n        \"samples\": [\n          768.0,\n          79.79947916666667,\n          127.25\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"BMI\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 262.05117817552093,\n        \"min\": 0.0,\n        \"max\": 768.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          31.992578124999998,\n          32.0,\n          768.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"DiabetesPedigreeFunction\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 271.3005221658502,\n        \"min\": 0.078,\n        \"max\": 768.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          0.47187630208333325,\n          0.3725,\n          768.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Age\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 260.1941178528413,\n        \"min\": 11.76023154067868,\n        \"max\": 768.0,\n        \"num_
```

```

unique_values\": 8,\n    \"samples\": [\n        33.240885416666664,\n        29.0,\n        768.0\n    ],\n    \"semantic_type\": \"\",\n    \"description\": \"\"\n  },\n  {\n    \"column\": \"Outcome\",\n    \"properties\": {\n      \"dtype\": \"number\",\n      \"std\": 271.3865920388932,\n      \"min\": 0.0,\n      \"max\": 768.0,\n      \"num_unique_values\": 5,\n      \"samples\": [\n        0.3489583333333333,\n        1.0,\n        0.4769513772427971\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    }\n  }\n]\n}\n},\n\"metadata\": {},\n\"execution_count\": 8\n}\n],\n},\n{\n  \"cell_type\": \"code\",\n  \"metadata\": {\n    \"colab\": {\n      \"base_uri\": \"https://localhost:8080/\",\n      \"height\": 178\n    },\n    \"id\": \"LrpHzaGpp5dQ\",\n    \"outputId\": \"a6b6e806-d36b-4f36-a9a8-f5ecd73d0be9\"\n  },\n  \"source\": [\n    \"diabetes_dataset['Outcome'].value_counts()\"\n  ],\n  \"execution_count\": 9,\n  \"outputs\": [\n    {\n      \"output_type\": \"execute_result\",\n      \"data\": {\n        \"text/plain\": [\n
```

```
"Outcome\n",
"0 500\n",
"1 268\n",
"Name: count, dtype: int64"
],
"text/html": [
"<div>\n",
"<style scoped>\n",
"  .dataframe tbody tr th:only-of-type {\n",
"    vertical-align: middle;\n",
"  }\n",
"\n",
"  .dataframe tbody tr th {\n",
"    vertical-align: top;\n",
"  }\n",
"\n",
"  .dataframe thead th {\n",
"    text-align: right;\n",
"  }\n",
"</style>\n",
"<table border='1' class='dataframe'>\n",
"  <thead>\n",
"    <tr style='text-align: right;'>\n",
"      <th></th>\n",
"      <th>count</th>\n",
"    </tr>\n",
"    <tr>\n",
"      <th>Outcome</th>\n",
"      <th></th>\n",
"    </tr>\n",
"  </thead>\n",
"  <tbody>\n",
"    <tr>\n",
```

```
"    <th>0</th>\n",
"    <td>500</td>\n",
"  </tr>\n",
"  <tr>\n",
"    <th>1</th>\n",
"    <td>268</td>\n",
"  </tr>\n",
" </tbody>\n",
"</table>\n",
"</div><br><label><b>dtype:</b> int64</label>"
]
},
"metadata": {},
"execution_count": 9
}
]
},
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  "metadata": {
    "id": "cB1qRaNcqeh5"
  },
  "source": [
    "0 --> Non-Diabetic\n",
    "\n",
    "1 --> Diabetic"
  ]
},
{
  "cell_type": "code",
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/",
```

```

"height": 143
},
"id": "I6MWR0k_qSCK",
"outputId": "41869107-49f6-4deb-a5a3-8bccb0bc9d00"
},
"source": [
  "diabetes_dataset.groupby('Outcome').mean()"
],
"execution_count": 10,
"outputs": [
  {
    "output_type": "execute_result",
    "data": {
      "text/plain": [
        "      Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin  \\\n",
        "Outcome                                         \n",
        "0      3.298000  109.980000   68.184000   19.664000  68.792000  \n",
        ",
        "1      4.865672  141.257463   70.824627   22.164179  100.335821  \n",
        "\n",
        "      BMI  DiabetesPedigreeFunction   Age  \n",
        "Outcome                                         \n",
        "0    30.304200          0.429734  31.190000  \n",
        "1    35.142537          0.550500  37.067164  "
      ],
      "text/html": [
        "\n",
        " <div id=\"df-8888dc2a-1145-4e57-bc62-355fdc159a34\" class=\"colab\n",
        "-df-container\">\n",
        "  <div>\n",
        "    <style scoped>\n",
        "      .dataframe tbody tr th:only-of-type {\n",
        "        vertical-align: middle;\n",

```

[illegible]

```
" </tr>\n",
" </thead>\n",
" <tbody>\n",
" <tr>\n",
" <th>0</th>\n",
" <td>3.298000</td>\n",
" <td>109.980000</td>\n",
" <td>68.184000</td>\n",
" <td>19.664000</td>\n",
" <td>68.792000</td>\n",
" <td>30.304200</td>\n",
" <td>0.429734</td>\n",
" <td>31.190000</td>\n",
" </tr>\n",
" <tr>\n",
" <th>1</th>\n",
" <td>4.865672</td>\n",
" <td>141.257463</td>\n",
" <td>70.824627</td>\n",
" <td>22.164179</td>\n",
" <td>100.335821</td>\n",
" <td>35.142537</td>\n",
" <td>0.550500</td>\n",
" <td>37.067164</td>\n",
" </tr>\n",
" </tbody>\n",
"</table>\n",
"</div>\n",
" <div class=\"colab-df-buttons\">\n",
"\n",
" <div class=\"colab-df-container\">\n",
" <button class=\"colab-df-convert\" onclick=\"convertToInteractive('df-888dc2a-1145-4e57-bc62-355fdc159a34')\">\n",
```

```

"      title=\"Convert this dataframe to an interactive table.\"\n",
"      style=\"display:none;\">\n",
"\n",
"    <svg xmlns=\"http://www.w3.org/2000/svg\" height=\"24px\" viewBox=
\"0 960 960 960\">\n",
"      <path d=\"M120 120v720h720v720H120Zm60 500h600v160H180v1
60Zm220 220h160v160H400v160Zm0 220h160v160H400v160ZM180 400h1
60v160H180v160Zm440 0h160v160H620v160ZM180 180h160v160H180v16
0Zm440 0h160v160H620v160Z\"/>\n",
"    </svg>\n",
"  </button>\n",
"\n",
"  <style>\n",
"    .colab-df-container {\n",
"      display: flex;\n",
"      gap: 12px;\n",
"    }\n",
"\n",
"    .colab-df-convert {\n",
"      background-color: #E8F0FE;\n",
"      border: none;\n",
"      border-radius: 50%;\n",
"      cursor: pointer;\n",
"      display: none;\n",
"      fill: #1967D2;\n",
"      height: 32px;\n",
"      padding: 0 0 0 0;\n",
"      width: 32px;\n",
"    }\n",
"\n",
"    .colab-df-convert:hover {\n",
"      background-color: #E2EBFA;\n",
"      box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(6

```

```
0, 64, 67, 0.15);\n",
    "    fill: #174EA6;\n",
    "  }\n",
    "\n",
    "  .colab-df-buttons div {\n",
    "    margin-bottom: 4px;\n",
    "  }\n",
    "\n",
    "  [theme=dark] .colab-df-convert {\n",
    "    background-color: #3B4455;\n",
    "    fill: #D2E3FC;\n",
    "  }\n",
    "\n",
    "  [theme=dark] .colab-df-convert:hover {\n",
    "    background-color: #434B5C;\n",
    "    box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);\n",
    "    filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));\n",
    "    fill: #FFFFFF;\n",
    "  }\n",
    "</style>\n",
    "\n",
    "<script>\n",
    "  const buttonEl =\n",
    "    document.querySelector('#df-8888dc2a-1145-4e57-bc62-355fdc15\n9a34 button.colab-df-convert');\n",
    "  buttonEl.style.display =\n",
    "    google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
    "\n",
    "  async function convertToInteractive(key) {\n",
    "    const element = document.querySelector('#df-8888dc2a-1145-4e57\n-bc62-355fdc159a34');\n",
    "    const dataTable =\n",
    "      await google.colab.kernel.invokeFunction('convertToInteractive',\n
```

```

"                [key], {});\n",
"    if (!dataTable) return;\n",
"\n",
"    const docLinkHtml = 'Like what you see? Visit the ' +\n",
"    '<a target=\"_blank\" href=https://colab.research.google.com/notebooks/data_table.ipynb>data table notebook</a>'\n",
"    + ' to learn more about interactive tables.';\n",
"    element.innerHTML = \";\n",
"    dataTable['output_type'] = 'display_data';\n",
"    await google.colab.output.renderOutput(dataTable, element);\n",
"    const docLink = document.createElement('div');\n",
"    docLink.innerHTML = docLinkHtml;\n",
"    element.appendChild(docLink);\n",
"  }\n",
" </script>\n",
" </div>\n",
"\n",
"\n",
" </div>\n",
" </div>\n"
],
"application/vnd.google.colaboratory.intrinsic+json": {
  "type": "dataframe",
  "summary": "{\n  \"name\": \"diabetes_dataset\",\n  \"rows\": 2,\n  \"fields\": [\n    {\n      \"column\": \"Outcome\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 0,\n        \"max\": 1,\n        \"num_unique_values\": 2,\n        \"samples\": [\n          1,\n          0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Pregnancies\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1.108511248584296,\n        \"min\": 3.298,\n        \"max\": 4.865671641791045,\n        \"num_unique_values\": 2,\n        \"samples\": [\n          4.865671641791045,\n          3.298\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Glucose\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 22.116505963980842,\n        \"min\":

```

```

109.98,\n    \"max\": 141.25746268656715,\n    \"num_unique_values\": 2,\n    \"samples\": [\n        141.25746268656715,\n        109.98\n    ],\n    \"semantic_type\": \"\",\n    \"description\": \"\"\n  }\n},\n {\n    \"column\": \"BloodPressure\",\n    \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1.8672051632998017,\n        \"min\": 68.184,\n        \"max\": 70.82462686567165,\n        \"num_unique_values\": 2,\n        \"samples\": [\n            70.82462686567165,\n            68.184\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n    }\n},\n {\n    \"column\": \"SkinThickness\",\n    \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1.7678935989570275,\n        \"min\": 19.664,\n        \"max\": 22.16417910447761,\n        \"num_unique_values\": 2,\n        \"samples\": [\n            22.16417910447761,\n            19.664\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n    }\n},\n {\n    \"column\": \"Insulin\",\n    \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 22.304849659757796,\n        \"min\": 68.792,\n        \"max\": 100.33582089552239,\n        \"num_unique_values\": 2,\n        \"samples\": [\n            100.33582089552239,\n            68.792\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n    }\n},\n {\n    \"column\": \"BMI\",\n    \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 3.4212211239962618,\n        \"min\": 30.3042,\n        \"max\": 35.14253731343284,\n        \"num_unique_values\": 2,\n        \"samples\": [\n            35.14253731343284,\n            30.3042\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n    }\n},\n {\n    \"column\": \"DiabetesPedigreeFunction\",\n    \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0.08539445753677459,\n        \"min\": 0.429734,\n        \"max\": 0.5505,\n        \"num_unique_values\": 2,\n        \"samples\": [\n            0.5505,\n            0.429734\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n    }\n},\n {\n    \"column\": \"Age\",\n    \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 4.155782645191446,\n        \"min\": 31.19,\n        \"max\": 37.06716417910448,\n        \"num_unique_values\": 2,\n        \"samples\": [\n            37.06716417910448,\n            31.19\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n    }\n  }\n]\n}\n},\n\"metadata\": {},\n\"execution_count\": 10

```

```
}  
]  
,  
{  
  "cell_type": "code",  
  "metadata": {  
    "id": "RoDW7l9mqqHZ"  
  },  
  "source": [  
    "# separating the data and labels\n",  
    "X = diabetes_dataset.drop(columns = 'Outcome', axis=1)\n",  
    "Y = diabetes_dataset['Outcome']"  
  ],  
  "execution_count": 11,  
  "outputs": []  
},  
{  
  "cell_type": "code",  
  "metadata": {  
    "colab": {  
      "base_uri": "https://localhost:8080/"  
    },  
    "id": "3eiRW9M9raMm",  
    "outputId": "c5555b36-2d02-40d4-f4b0-90a5ac86d780"  
  },  
  "source": [  
    "print(X)"  
  ],  
  "execution_count": 12,  
  "outputs": [  
    {  
      "output_type": "stream",  
      "name": "stdout",
```

```

"text": [
  "  Pregnancies Glucose BloodPressure SkinThickness Insulin  BMI  \\n",
  "0      6   148      72      35    0 33.6  \\n",
  "1      1    85      66      29    0 26.6  \\n",
  "2      8   183      64       0    0 23.3  \\n",
  "3      1    89      66      23   94 28.1  \\n",
  "4      0   137      40      35   168 43.1  \\n",
  "..     ...   ...      ...      ...   ...  \\n",
  "763     10   101      76      48   180 32.9  \\n",
  "764      2   122      70      27    0 36.8  \\n",
  "765      5   121      72      23   112 26.2  \\n",
  "766      1   126      60       0    0 30.1  \\n",
  "767      1    93      70      31    0 30.4  \\n",
  "\\n",
  "  DiabetesPedigreeFunction Age  \\n",
  "0          0.627  50  \\n",
  "1          0.351  31  \\n",
  "2          0.672  32  \\n",
  "3          0.167  21  \\n",
  "4          2.288  33  \\n",
  "..         ...  ...  \\n",
  "763         0.171  63  \\n",
  "764         0.340  27  \\n",
  "765         0.245  30  \\n",
  "766         0.349  47  \\n",
  "767         0.315  23  \\n",
  "\\n",
  "[768 rows x 8 columns]\\n"
]
}
]
},
{

```

```
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  },
  "id": "AoxgTJAMrcCI",
  "outputId": "e2165d22-eaad-4525-8639-66ac5931108b"
},
"source": [
  "print(Y)"
],
"execution_count": 13,
"outputs": [
  {
    "output_type": "stream",
    "name": "stdout",
    "text": [
      "0  1\n",
      "1  0\n",
      "2  1\n",
      "3  0\n",
      "4  1\n",
      "  ..\n",
      "763  0\n",
      "764  0\n",
      "765  0\n",
      "766  1\n",
      "767  0\n",
      "Name: Outcome, Length: 768, dtype: int64\n"
    ]
  }
],
},
```

```
{
  "cell_type": "markdown",
  "metadata": {
    "id": "umAbo_kqrlzl"
  },
  "source": [
    "Data Standardization"
  ]
},
{
  "cell_type": "code",
  "metadata": {
    "id": "njfM5X60rgnc"
  },
  "source": [
    "scaler = StandardScaler()"
  ],
  "execution_count": 14,
  "outputs": []
},
{
  "cell_type": "code",
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/",
      "height": 80
    },
    "id": "g0ai5ARbr53p",
    "outputId": "106075ea-9c21-457f-8c0a-b4b47a268026"
  },
  "source": [
    "scaler.fit(X)"
  ],
}
```

```
"execution_count": 15,
"outputs": [
  {
    "output_type": "execute_result",
    "data": {
      "text/plain": [
        "StandardScaler()"
      ],
      "text/html": [
        "<style>#sk-container-id-1 {\n",
        " /* Definition of color scheme common for light and dark mode */\n",
        " --sklearn-color-text: #000;\n",
        " --sklearn-color-text-muted: #666;\n",
        " --sklearn-color-line: gray;\n",
        " /* Definition of color scheme for unfitted estimators */\n",
        " --sklearn-color-unfitted-level-0: #fff5e6;\n",
        " --sklearn-color-unfitted-level-1: #f6e4d2;\n",
        " --sklearn-color-unfitted-level-2: #ffe0b3;\n",
        " --sklearn-color-unfitted-level-3: chocolate;\n",
        " /* Definition of color scheme for fitted estimators */\n",
        " --sklearn-color-fitted-level-0: #f0f8ff;\n",
        " --sklearn-color-fitted-level-1: #d4ebff;\n",
        " --sklearn-color-fitted-level-2: #b3dbfd;\n",
        " --sklearn-color-fitted-level-3: cornflowerblue;\n",
        "\n",
        " /* Specific color for light theme */\n",
        " --sklearn-color-text-on-default-background: var(--sg-text-color, var(--t",
        "heme-code-foreground, var(--jp-content-font-color1, black))); \n",
        " --sklearn-color-background: var(--sg-background-color, var(--theme-b",
        "ackground, var(--jp-layout-color0, white))); \n",
        " --sklearn-color-border-box: var(--sg-text-color, var(--theme-code-fore",
        "ground, var(--jp-content-font-color1, black))); \n",
        " --sklearn-color-icon: #696969;\n",
```

```
"\n",
" @media (prefers-color-scheme: dark) {\n",
" /* Redefinition of color scheme for dark theme */\n",
" --sklearn-color-text-on-default-background: var(--sg-text-color, var(--theme-code-foreground, var(--jp-content-font-color1, white))); \n",
" --sklearn-color-background: var(--sg-background-color, var(--theme-background, var(--jp-layout-color0, #111))); \n",
" --sklearn-color-border-box: var(--sg-text-color, var(--theme-code-foreground, var(--jp-content-font-color1, white))); \n",
" --sklearn-color-icon: #878787; \n",
" }\n",
"}\n",
"\n",
"#sk-container-id-1 {\n",
" color: var(--sklearn-color-text); \n",
"}\n",
"\n",
"#sk-container-id-1 pre {\n",
" padding: 0; \n",
"}\n",
"\n",
"#sk-container-id-1 input.sk-hidden--visually {\n",
" border: 0; \n",
" clip: rect(1px 1px 1px 1px); \n",
" clip: rect(1px, 1px, 1px, 1px); \n",
" height: 1px; \n",
" margin: -1px; \n",
" overflow: hidden; \n",
" padding: 0; \n",
" position: absolute; \n",
" width: 1px; \n",
"}\n",
"\n",
```

```
"#sk-container-id-1 div.sk-dashed-wrapped {\n",\n" border: 1px dashed var(--sklearn-color-line);\n",\n" margin: 0 0.4em 0.5em 0.4em;\n",\n" box-sizing: border-box;\n",\n" padding-bottom: 0.4em;\n",\n" background-color: var(--sklearn-color-background);\n",\n"}\n",\n"\n",\n"#sk-container-id-1 div.sk-container {\n",\n" /* jupyter's `normalize.less` sets `[hidden] { display: none; }`\n",\n" but bootstrap.min.css set `[hidden] { display: none !important; }`\n",\n" so we also need the `!important` here to be able to override the\n",\n" default hidden behavior on the sphinx rendered scikit-learn.org.\n",\n" See: https://github.com/scikit-learn/scikit-learn/issues/21755 */\n",\n" display: inline-block !important;\n",\n" position: relative;\n",\n"}\n",\n"\n",\n"#sk-container-id-1 div.sk-text-repr-fallback {\n",\n" display: none;\n",\n"}\n",\n"\n",\n"div.sk-parallel-item,\n",\n"div.sk-serial,\n",\n"div.sk-item {\n",\n" /* draw centered vertical line to link estimators */\n",\n" background-image: linear-gradient(var(--sklearn-color-text-on-default-\nbackground), var(--sklearn-color-text-on-default-background));\n",\n" background-size: 2px 100%;\n",\n" background-repeat: no-repeat;\n",\n" background-position: center center;\n",\n"}\n",\n"\n",
```

```
"/* Parallel-specific style estimator block */\n",
"\n",
"#sk-container-id-1 div.sk-parallel-item::after {\n",
"  content: \"\";\n",
"  width: 100%;\n",
"  border-bottom: 2px solid var(--sklearn-color-text-on-default-backgroun\n",
d);\n",
"  flex-grow: 1;\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-parallel {\n",
"  display: flex;\n",
"  align-items: stretch;\n",
"  justify-content: center;\n",
"  background-color: var(--sklearn-color-background);\n",
"  position: relative;\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-parallel-item {\n",
"  display: flex;\n",
"  flex-direction: column;\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-parallel-item:first-child::after {\n",
"  align-self: flex-end;\n",
"  width: 50%;\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-parallel-item:last-child::after {\n",
"  align-self: flex-start;\n",
"  width: 50%;\n",
"}\n",
"\n",
```

```

"#sk-container-id-1 div.sk-parallel-item:only-child::after {\n",
" width: 0;\n",
"}\n",
"\n",
"/* Serial-specific style estimator block */\n",
"\n",
"#sk-container-id-1 div.sk-serial {\n",
" display: flex;\n",
" flex-direction: column;\n",
" align-items: center;\n",
" background-color: var(--sklearn-color-background);\n",
" padding-right: 1em;\n",
" padding-left: 1em;\n",
"}\n",
"\n",
"\n",
"/* Toggleable style: style used for estimator/Pipeline/ColumnTransformer
box that is\n",
"clickable and can be expanded/collapsed.\n",
"- Pipeline and ColumnTransformer use this feature and define the default st
yle\n",
"- Estimators will overwrite some part of the style using the `sk-estimator`
class\n",
"*/\n",
"\n",
"/* Pipeline and ColumnTransformer style (default) */\n",
"\n",
"#sk-container-id-1 div.sk-toggleable {\n",
" /* Default theme specific background. It is overwritten whether we have a
\n",
" specific estimator or a Pipeline/ColumnTransformer */\n",
" background-color: var(--sklearn-color-background);\n",
"}\n",

```

```
"\n",
"/ * Toggleable label */\n",
"#sk-container-id-1 label.sk-toggleable__label {\n",
"  cursor: pointer;\n",
"  display: flex;\n",
"  width: 100%;\n",
"  margin-bottom: 0;\n",
"  padding: 0.5em;\n",
"  box-sizing: border-box;\n",
"  text-align: center;\n",
"  align-items: start;\n",
"  justify-content: space-between;\n",
"  gap: 0.5em;\n",
"}\n",
"\n",
"#sk-container-id-1 label.sk-toggleable__label.caption {\n",
"  font-size: 0.6rem;\n",
"  font-weight: lighter;\n",
"  color: var(--sklearn-color-text-muted);\n",
"}\n",
"\n",
"#sk-container-id-1 label.sk-toggleable__label-arrow:before {\n",
"  /* Arrow on the left of the label */\n",
"  content: \">\";\n",
"  float: left;\n",
"  margin-right: 0.25em;\n",
"  color: var(--sklearn-color-icon);\n",
"}\n",
"\n",
"#sk-container-id-1 label.sk-toggleable__label-arrow:hover:before {\n",
"  color: var(--sklearn-color-text);\n",
"}\n",
"\n",
```

```
/* Toggleable content - dropdown */\n",
"\n",
"#sk-container-id-1 div.sk-toggleable__content {\n",
"  max-height: 0;\n",
"  max-width: 0;\n",
"  overflow: hidden;\n",
"  text-align: left;\n",
"  /* unfitted */\n",
"  background-color: var(--sklearn-color-unfitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-toggleable__content.fitted {\n",
"  /* fitted */\n",
"  background-color: var(--sklearn-color-fitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-toggleable__content pre {\n",
"  margin: 0.2em;\n",
"  border-radius: 0.25em;\n",
"  color: var(--sklearn-color-text);\n",
"  /* unfitted */\n",
"  background-color: var(--sklearn-color-unfitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-toggleable__content.fitted pre {\n",
"  /* unfitted */\n",
"  background-color: var(--sklearn-color-fitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-1 input.sk-toggleable__control:checked~div.sk-toggleable__content {\n",
"  /* Expand drop-down */\n",
"  max-height: 200px;\n",
}
```

```

" max-width: 100%;\n",
" overflow: auto;\n",
"}\n",
"\n",
"#sk-container-id-1 input.sk-toggleable__control:checked~label.sk-toggleable__label-arrow:before {\n",
" content: \"▼\";\n",
"}\n",
"\n",
"/* Pipeline/ColumnTransformer-specific style */\n",
"\n",
"#sk-container-id-1 div.sk-label input.sk-toggleable__control:checked~label.sk-toggleable__label {\n",
" color: var(--sklearn-color-text);\n",
" background-color: var(--sklearn-color-unfitted-level-2);\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-label.fitted input.sk-toggleable__control:checked~label.sk-toggleable__label {\n",
" background-color: var(--sklearn-color-fitted-level-2);\n",
"}\n",
"\n",
"/* Estimator-specific style */\n",
"\n",
"/* Colorize estimator box */\n",
"#sk-container-id-1 div.sk-estimator input.sk-toggleable__control:checked~label.sk-toggleable__label {\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-2);\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-estimator.fitted input.sk-toggleable__control:checked~label.sk-toggleable__label {\n",

```

```

" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-2);\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-label label.sk-toggleable__label,\n",
"#sk-container-id-1 div.sk-label label {\n",
" /* The background is the default theme color */\n",
" color: var(--sklearn-color-text-on-default-background);\n",
"}\n",
"\n",
"/* On hover, darken the color of the background */\n",
"#sk-container-id-1 div.sk-label:hover label.sk-toggleable__label {\n",
" color: var(--sklearn-color-text);\n",
" background-color: var(--sklearn-color-unfitted-level-2);\n",
"}\n",
"\n",
"/* Label box, darken color on hover, fitted */\n",
"#sk-container-id-1 div.sk-label.fitted:hover label.sk-toggleable__label.fitt
ed {\n",
" color: var(--sklearn-color-text);\n",
" background-color: var(--sklearn-color-fitted-level-2);\n",
"}\n",
"\n",
"/* Estimator label */\n",
"\n",
"#sk-container-id-1 div.sk-label label {\n",
" font-family: monospace;\n",
" font-weight: bold;\n",
" display: inline-block;\n",
" line-height: 1.2em;\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-label-container {\n",

```

```
" text-align: center;\n",
"}\n",
"\n",
"/* Estimator-specific */\n",
"#sk-container-id-1 div.sk-estimator {\n",
" font-family: monospace;\n",
" border: 1px dotted var(--sklearn-color-border-box);\n",
" border-radius: 0.25em;\n",
" box-sizing: border-box;\n",
" margin-bottom: 0.5em;\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-estimator.fitted {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-0);\n",
"}\n",
"\n",
"/* on hover */\n",
"#sk-container-id-1 div.sk-estimator:hover {\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-2);\n",
"}\n",
"\n",
"#sk-container-id-1 div.sk-estimator.fitted:hover {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-2);\n",
"}\n",
"\n",
"/* Specification for estimator info (e.g. \"i\" and \"?\") */\n",
"\n",
"/* Common style for \"i\" and \"?\" */\n",
```

```
"\n",
".sk-estimator-doc-link,\n",
"a:link.sk-estimator-doc-link,\n",
"a:visited.sk-estimator-doc-link {\n",
"  float: right;\n",
"  font-size: smaller;\n",
"  line-height: 1em;\n",
"  font-family: monospace;\n",
"  background-color: var(--sklearn-color-background);\n",
"  border-radius: 1em;\n",
"  height: 1em;\n",
"  width: 1em;\n",
"  text-decoration: none !important;\n",
"  margin-left: 0.5em;\n",
"  text-align: center;\n",
"  /* unfitted */\n",
"  border: var(--sklearn-color-unfitted-level-1) 1pt solid;\n",
"  color: var(--sklearn-color-unfitted-level-1);\n",
"}\n",
"\n",
".sk-estimator-doc-link.fitted,\n",
"a:link.sk-estimator-doc-link.fitted,\n",
"a:visited.sk-estimator-doc-link.fitted {\n",
"  /* fitted */\n",
"  border: var(--sklearn-color-fitted-level-1) 1pt solid;\n",
"  color: var(--sklearn-color-fitted-level-1);\n",
"}\n",
"\n",
"/* On hover */\n",
"div.sk-estimator:hover .sk-estimator-doc-link:hover,\n",
".sk-estimator-doc-link:hover,\n",
"div.sk-label-container:hover .sk-estimator-doc-link:hover,\n",
".sk-estimator-doc-link:hover {\n",
```

```
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-3);\n",
" color: var(--sklearn-color-background);\n",
" text-decoration: none;\n",
"}\n",
"\n",
"div.sk-estimator.fitted: hover .sk-estimator-doc-link.fitted: hover,\n",
".sk-estimator-doc-link.fitted: hover,\n",
"div.sk-label-container: hover .sk-estimator-doc-link.fitted: hover,\n",
".sk-estimator-doc-link.fitted: hover {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-3);\n",
" color: var(--sklearn-color-background);\n",
" text-decoration: none;\n",
"}\n",
"\n",
"/* Span, style for the box shown on hovering the info icon */\n",
".sk-estimator-doc-link span {\n",
" display: none;\n",
" z-index: 9999;\n",
" position: relative;\n",
" font-weight: normal;\n",
" right: .2ex;\n",
" padding: .5ex;\n",
" margin: .5ex;\n",
" width: min-content;\n",
" min-width: 20ex;\n",
" max-width: 50ex;\n",
" color: var(--sklearn-color-text);\n",
" box-shadow: 2pt 2pt 4pt #999;\n",
" /* unfitted */\n",
" background: var(--sklearn-color-unfitted-level-0);\n",
" border: .5pt solid var(--sklearn-color-unfitted-level-3);\n",
```

```
"}\n",
"\n",
".sk-estimator-doc-link.fitted span {\n",
" /* fitted */\n",
" background: var(--sklearn-color-fitted-level-0);\n",
" border: var(--sklearn-color-fitted-level-3);\n",
"}\n",
"\n",
".sk-estimator-doc-link:hover span {\n",
" display: block;\n",
"}\n",
"\n",
"/* \"?\"-specific style due to the `<a>` HTML tag */\n",
"\n",
"#sk-container-id-1 a.estimator_doc_link {\n",
" float: right;\n",
" font-size: 1rem;\n",
" line-height: 1em;\n",
" font-family: monospace;\n",
" background-color: var(--sklearn-color-background);\n",
" border-radius: 1rem;\n",
" height: 1rem;\n",
" width: 1rem;\n",
" text-decoration: none;\n",
" /* unfitted */\n",
" color: var(--sklearn-color-unfitted-level-1);\n",
" border: var(--sklearn-color-unfitted-level-1) 1pt solid;\n",
"}\n",
"\n",
"#sk-container-id-1 a.estimator_doc_link.fitted {\n",
" /* fitted */\n",
" border: var(--sklearn-color-fitted-level-1) 1pt solid;\n",
" color: var(--sklearn-color-fitted-level-1);\n",
}
```

"}\n",

"\n",

```
"/* On hover */\n",
```

```
"#sk-container-id-1 a.estimator_doc_link:hover {\n"
```

```
" /* unfitted */\n",
```

```
" background-color: var(--sklearn-color-unfitted-level-3);\n",
```

```
" color: var(--sklearn-color-background);\n",
```

" text-decoration: none;\n",

"}\n",

"\n",

```
"#sk-container-id-1 a.estimator_doc_link.fitted: hover {\n",
```

```
" /* fitted */\n",
```

```
" background-color: var(--sklearn-color-fitted-level-3);\n",
```

"}\n",

```
</style><div id=\"sk-container-id-1\" class=\"sk-top-container\"><div class=
s=\"sk-text-repr-fallback\"><pre>StandardScaler()</pre><b>In a Jupyter enviro
nment, please rerun this cell to show the HTML representation or trust the notebo
ok. <br />On GitHub, the HTML representation is unable to render, please try loadi
ng this page with nbviewer.org.</b></div><div class=\"sk-container\" hidden><d
iv class=\"sk-item\"><div class=\"sk-estimator fitted sk-toggleable\"><input class
=\"sk-toggleable__control sk-hidden--visually\" id=\"sk-estimator-id-1\" type=\"c
heckbox\" checked><label for=\"sk-estimator-id-1\" class=\"sk-toggleable__label
fitted sk-toggleable__label-arrow\"><div><div>StandardScaler</div></div><di
v><a class=\"sk-estimator-doc-link fitted\" rel=\"noreferrer\" target=\"_blank\" hre
f=\"https://scikit-learn.org/1.6/modules/generated/sklearn.preprocessing.Standa
rdScaler.html\">?<span>Documentation for StandardScaler</span></a><span c
lass=\"sk-estimator-doc-link fitted\">i<span>Fitted</span></span></div></labe
l><div class=\"sk-toggleable__content fitted\"><pre>StandardScaler()</pre></
div> </div></div></div></div>
```

]

 $\},$

```
"metadata": {},
```

```
"execution_count": 15
```

```
}
]
},
{
  "cell_type": "code",
  "metadata": {
    "id": "FHxNwPuZr-kD"
  },
  "source": [
    "standardized_data = scaler.transform(X)"
  ],
  "execution_count": 16,
  "outputs": []
},
{
  "cell_type": "code",
  "metadata": {
    "colab": {
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    },
    "id": "fjMwZ5x6sPUJ",
    "outputId": "ac8a4fdb-2f30-47d2-cef8-2147498c5826"
  },
  "source": [
    "print(standardized_data)"
  ],
  "execution_count": 17,
  "outputs": [
    {
      "output_type": "stream",
      "name": "stdout",
      "text": [
        "[[ 0.63994726  0.84832379  0.14964075 ...  0.20401277  0.46849198\n",
```

```

" 1.4259954 ]\n",
" [-0.84488505 -1.12339636 -0.16054575 ... -0.68442195 -0.36506078\n"
,
" -0.19067191]\n",
" [ 1.23388019 1.94372388 -0.26394125 ... -1.10325546 0.60439732\n",
" -0.10558415]\n",
" ...\n",
" [ 0.3429808 0.00330087 0.14964075 ... -0.73518964 -0.68519336\n",
" -0.27575966]\n",
" [-0.84488505 0.1597866 -0.47073225 ... -0.24020459 -0.37110101\n",
" 1.17073215]\n",
" [-0.84488505 -0.8730192 0.04624525 ... -0.20212881 -0.47378505\n",
" -0.87137393]]\n"
]
}
]
},
{
"cell_type": "code",
"metadata": {
"id": "ZxWSl4SGsRjE"
},
"source": [
"X = standardized_data\n",
"Y = diabetes_dataset['Outcome']"
],
"execution_count": 18,
"outputs": []
},
{
"cell_type": "code",
"metadata": {
"colab": {

```

```
"base_uri": "https://localhost:8080/"
},
"id": "lhJF_7QjsjmP",
"outputId": "b955de53-8857-47ee-f78a-e9c0ce54b217"
},
"source": [
  "print(X)\n",
  "print(Y)"
],
"execution_count": 19,
"outputs": [
  {
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    "name": "stdout",
    "text": [
      "[[ 0.63994726  0.84832379  0.14964075 ...  0.20401277  0.46849198\n",
      "  1.4259954 ]\n",
      " [-0.84488505 -1.12339636 -0.16054575 ... -0.68442195 -0.36506078\n",
      "  -0.19067191]\n",
      " [ 1.23388019  1.94372388 -0.26394125 ... -1.10325546  0.60439732\n",
      "  -0.10558415]\n",
      " ...\n",
      " [ 0.3429808  0.00330087  0.14964075 ... -0.73518964 -0.68519336\n",
      "  -0.27575966]\n",
      " [-0.84488505  0.1597866  -0.47073225 ... -0.24020459 -0.37110101\n",
      "  1.17073215]\n",
      " [-0.84488505 -0.8730192  0.04624525 ... -0.20212881 -0.47378505\n",
      "  -0.87137393]]\n",
      "0  1\n",
      "1  0\n",
      "2  1\n",
      "3  0
```

```

"4  1\n",
"  ..\n",
"763  0\n",
"764  0\n",
"765  0\n",
"766  1\n",
"767  0\n",
"Name: Outcome, Length: 768, dtype: int64\n"
]
}
]
},
{
"cell_type": "markdown",
"metadata": {
"id": "gHciEFkxsoQP"
},
"source": [
"Train Test Split"
]
},
{
"cell_type": "code",
"metadata": {
"id": "AEfKGj_yslvD"
},
"source": [
"X_train, X_test, Y_train, Y_test = train_test_split(X,Y, test_size = 0.2, stratify
=Y, random_state=2)"
],
"execution_count": 20,
"outputs": []
},

```

```
{
  "cell_type": "code",
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    "colab": {
      "base_uri": "https://localhost:8080/"
    },
    "id": "DR05T-o0t3FQ",
    "outputId": "6195f2c9-be46-4d8e-9b7b-ccb8d7c8b34e"
  },
  "source": [
    "print(X.shape, X_train.shape, X_test.shape)"
  ],
  "execution_count": 21,
  "outputs": [
    {
      "output_type": "stream",
      "name": "stdout",
      "text": [
        "(768, 8) (614, 8) (154, 8)\n"
      ]
    }
  ],
},
{
  "cell_type": "markdown",
  "metadata": {
    "id": "EIJ3tkOtuC_n"
  },
  "source": [
    "Training the Model"
  ],
},
{
```

```
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"metadata": {
  "id": "5szLWHINt9xc"
},
"source": [
  "classifier = svm.SVC(kernel='linear')"
],
"execution_count": 22,
"outputs": []
},
{
  "cell_type": "code",
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/",
      "height": 80
    },
    "id": "ncJWY_7suPAb",
    "outputId": "67c7b9de-2d32-4db2-f5e7-92c2c3e0295b"
  },
  "source": [
    "#training the support vector Machine Classifier\n",
    "classifier.fit(X_train, Y_train)"
  ],
  "execution_count": 23,
  "outputs": [
    {
      "output_type": "execute_result",
      "data": {
        "text/plain": [
          "SVC(kernel='linear')"
        ],
        "text/html": [
```

```

"<style>#sk-container-id-2 {\n",
" /* Definition of color scheme common for light and dark mode */\n",
" --sklearn-color-text: #000;\n",
" --sklearn-color-text-muted: #666;\n",
" --sklearn-color-line: gray;\n",
" /* Definition of color scheme for unfitted estimators */\n",
" --sklearn-color-unfitted-level-0: #fff5e6;\n",
" --sklearn-color-unfitted-level-1: #f6e4d2;\n",
" --sklearn-color-unfitted-level-2: #ffe0b3;\n",
" --sklearn-color-unfitted-level-3: chocolate;\n",
" /* Definition of color scheme for fitted estimators */\n",
" --sklearn-color-fitted-level-0: #f0f8ff;\n",
" --sklearn-color-fitted-level-1: #d4ebff;\n",
" --sklearn-color-fitted-level-2: #b3dbfd;\n",
" --sklearn-color-fitted-level-3: cornflowerblue;\n",
"\n",
" /* Specific color for light theme */\n",
" --sklearn-color-text-on-default-background: var(--sg-text-color, var(--t
heme-code-foreground, var(--jp-content-font-color1, black))); \n",
" --sklearn-color-background: var(--sg-background-color, var(--theme-b
ackground, var(--jp-layout-color0, white))); \n",
" --sklearn-color-border-box: var(--sg-text-color, var(--theme-code-fore
ground, var(--jp-content-font-color1, black))); \n",
" --sklearn-color-icon: #696969;\n",
"\n",
" @media (prefers-color-scheme: dark) {\n",
" /* Redefinition of color scheme for dark theme */\n",
" --sklearn-color-text-on-default-background: var(--sg-text-color, var(--t
heme-code-foreground, var(--jp-content-font-color1, white))); \n",
" --sklearn-color-background: var(--sg-background-color, var(--theme-b
ackground, var(--jp-layout-color0, #111))); \n",
" --sklearn-color-border-box: var(--sg-text-color, var(--theme-code-fore
ground, var(--jp-content-font-color1, white))); \n",

```

```
"  --sklearn-color-icon: #878787;\n",\n"}\n",\n"}\n",\n"\n",\n"#sk-container-id-2 {\n",\n"  color: var(--sklearn-color-text);\n",\n"}\n",\n"\n",\n"#sk-container-id-2 pre {\n",\n"  padding: 0;\n",\n"}\n",\n"\n",\n"#sk-container-id-2 input.sk-hidden--visually {\n",\n"  border: 0;\n",\n"  clip: rect(1px 1px 1px 1px);\n",\n"  clip: rect(1px, 1px, 1px, 1px);\n",\n"  height: 1px;\n",\n"  margin: -1px;\n",\n"  overflow: hidden;\n",\n"  padding: 0;\n",\n"  position: absolute;\n",\n"  width: 1px;\n",\n"}\n",\n"\n",\n"#sk-container-id-2 div.sk-dashed-wrapped {\n",\n"  border: 1px dashed var(--sklearn-color-line);\n",\n"  margin: 0 0.4em 0.5em 0.4em;\n",\n"  box-sizing: border-box;\n",\n"  padding-bottom: 0.4em;\n",\n"  background-color: var(--sklearn-color-background);\n",\n"}\n",\n"\n",\n"#sk-container-id-2 div.sk-container {\n",
```

```
" /* jupyter's `normalize.less` sets `[hidden] { display: none; }` \n",
"   but bootstrap.min.css set `[hidden] { display: none !important; }` \n",
"   so we also need the `!important` here to be able to override the\n",
"   default hidden behavior on the sphinx rendered scikit-learn.org.\n",
"   See: https://github.com/scikit-learn/scikit-learn/issues/21755 */\n",
" display: inline-block !important;\n",
" position: relative;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-text-repr-fallback {\n",
" display: none;\n",
"}\n",
"\n",
"div.sk-parallel-item,\n",
"div.sk-serial,\n",
"div.sk-item {\n",
" /* draw centered vertical line to link estimators */\n",
" background-image: linear-gradient(var(--sklearn-color-text-on-default-
background), var(--sklearn-color-text-on-default-background));\n",
" background-size: 2px 100%;\n",
" background-repeat: no-repeat;\n",
" background-position: center center;\n",
"}\n",
"\n",
"/* Parallel-specific style estimator block */\n",
"\n",
"#sk-container-id-2 div.sk-parallel-item::after {\n",
" content: \"\";\n",
" width: 100%;\n",
" border-bottom: 2px solid var(--sklearn-color-text-on-default-backgroun
d);\n",
" flex-grow: 1;\n",
"}\n",
```

```
"\n",
"#sk-container-id-2 div.sk-parallel {\n",
" display: flex;\n",
" align-items: stretch;\n",
" justify-content: center;\n",
" background-color: var(--sklearn-color-background);\n",
" position: relative;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-parallel-item {\n",
" display: flex;\n",
" flex-direction: column;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-parallel-item:first-child::after {\n",
" align-self: flex-end;\n",
" width: 50%;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-parallel-item:last-child::after {\n",
" align-self: flex-start;\n",
" width: 50%;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-parallel-item:only-child::after {\n",
" width: 0;\n",
"}\n",
"\n",
/* Serial-specific style estimator block */\n",
"\n",
"#sk-container-id-2 div.sk-serial {\n",
" display: flex;\n",
" flex-direction: column;
```

```

" align-items: center;\n",
" background-color: var(--sklearn-color-background);\n",
" padding-right: 1em;\n",
" padding-left: 1em;\n",
"}\n",
"\n",
"\n",
/* Toggleable style: style used for estimator/Pipeline/ColumnTransformer
box that is\n",
"clickable and can be expanded/collapsed.\n",
"- Pipeline and ColumnTransformer use this feature and define the default st
yle\n",
"- Estimators will overwrite some part of the style using the `sk-estimator`
class\n",
"*/\n",
"\n",
/* Pipeline and ColumnTransformer style (default) */\n",
"\n",
"#sk-container-id-2 div.sk-toggleable {\n",
" /* Default theme specific background. It is overwritten whether we have a
\n",
" specific estimator or a Pipeline/ColumnTransformer */\n",
" background-color: var(--sklearn-color-background);\n",
"}\n",
"\n",
/* Toggleable label */\n",
"#sk-container-id-2 label.sk-toggleable__label {\n",
" cursor: pointer;\n",
" display: flex;\n",
" width: 100%;\n",
" margin-bottom: 0;\n",
" padding: 0.5em;\n",
" box-sizing: border-box;\n",

```

```
" text-align: center;\n",
" align-items: start;\n",
" justify-content: space-between;\n",
" gap: 0.5em;\n",
"}\n",
"\n",
"#sk-container-id-2 label.sk-toggleable__label.caption {\n",
" font-size: 0.6rem;\n",
" font-weight: lighter;\n",
" color: var(--sklearn-color-text-muted);\n",
"}\n",
"\n",
"#sk-container-id-2 label.sk-toggleable__label-arrow:before {\n",
" /* Arrow on the left of the label */\n",
" content: \">\";\n",
" float: left;\n",
" margin-right: 0.25em;\n",
" color: var(--sklearn-color-icon);\n",
"}\n",
"\n",
"#sk-container-id-2 label.sk-toggleable__label-arrow:hover:before {\n",
" color: var(--sklearn-color-text);\n",
"}\n",
"\n",
"/* Toggleable content - dropdown */\n",
"\n",
"#sk-container-id-2 div.sk-toggleable__content {\n",
" max-height: 0;\n",
" max-width: 0;\n",
" overflow: hidden;\n",
" text-align: left;\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-0);\n",
```

```
"}\n",
"\n",
"#sk-container-id-2 div.sk-toggleable__content.fitted {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-toggleable__content pre {\n",
" margin: 0.2em;\n",
" border-radius: 0.25em;\n",
" color: var(--sklearn-color-text);\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-toggleable__content.fitted pre {\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-fitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-2 input.sk-toggleable__control:checked~div.sk-toggleable__content {\n",
" /* Expand drop-down */\n",
" max-height: 200px;\n",
" max-width: 100%;\n",
" overflow: auto;\n",
"}\n",
"\n",
"#sk-container-id-2 input.sk-toggleable__control:checked~label.sk-toggleable__label-arrow:before {\n",
" content: \"▼\";\n",
"}\n",
"\n",
```

```

/* Pipeline/ColumnTransformer-specific style */\n",
"\n",
"#sk-container-id-2 div.sk-label input.sk-toggleable__control:checked~label.sk-toggleable__label {\n",
"  color: var(--sklearn-color-text);\n",
"  background-color: var(--sklearn-color-unfitted-level-2);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-label.fitted input.sk-toggleable__control:checked~label.sk-toggleable__label {\n",
"  background-color: var(--sklearn-color-fitted-level-2);\n",
"}\n",
"\n",
/* Estimator-specific style */\n",
"\n",
/* Colorize estimator box */\n",
"#sk-container-id-2 div.sk-estimator input.sk-toggleable__control:checked~label.sk-toggleable__label {\n",
"  /* unfitted */\n",
"  background-color: var(--sklearn-color-unfitted-level-2);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-estimator.fitted input.sk-toggleable__control:checked~label.sk-toggleable__label {\n",
"  /* fitted */\n",
"  background-color: var(--sklearn-color-fitted-level-2);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-label label.sk-toggleable__label,\n",
"#sk-container-id-2 div.sk-label label {\n",
"  /* The background is the default theme color */\n",
"  color: var(--sklearn-color-text-on-default-background);\n",
"}\n",

```

```
"\n",
"/* On hover, darken the color of the background */\n",
"#sk-container-id-2 div.sk-label:hover label.sk-toggleable__label {\n",
"  color: var(--sklearn-color-text);\n",
"  background-color: var(--sklearn-color-unfitted-level-2);\n",
"}\n",
"\n",
"/* Label box, darken color on hover, fitted */\n",
"#sk-container-id-2 div.sk-label.fitted:hover label.sk-toggleable__label.fitted {\n",
"  color: var(--sklearn-color-text);\n",
"  background-color: var(--sklearn-color-fitted-level-2);\n",
"}\n",
"\n",
"/* Estimator label */\n",
"\n",
"#sk-container-id-2 div.sk-label label {\n",
"  font-family: monospace;\n",
"  font-weight: bold;\n",
"  display: inline-block;\n",
"  line-height: 1.2em;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-label-container {\n",
"  text-align: center;\n",
"}\n",
"\n",
"/* Estimator-specific */\n",
"#sk-container-id-2 div.sk-estimator {\n",
"  font-family: monospace;\n",
"  border: 1px dotted var(--sklearn-color-border-box);\n",
"  border-radius: 0.25em;\n",
"  box-sizing: border-box;\n",
}
```

```
" margin-bottom: 0.5em;\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-estimator.fitted {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-0);\n",
"}\n",
"\n",
"/* on hover */\n",
"#sk-container-id-2 div.sk-estimator:hover {\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-2);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-estimator.fitted:hover {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-2);\n",
"}\n",
"\n",
"/* Specification for estimator info (e.g. \"i\" and \"?\") */\n",
"\n",
"/* Common style for \"i\" and \"?\" */\n",
"\n",
".sk-estimator-doc-link,\n",
"a:link.sk-estimator-doc-link,\n",
"a:visited.sk-estimator-doc-link {\n",
" float: right;\n",
" font-size: smaller;\n",
" line-height: 1em;\n",
" font-family: monospace;\n",
" background-color: var(--sklearn-color-background);\n",
```

```
" border-radius: 1em;\n",
" height: 1em;\n",
" width: 1em;\n",
" text-decoration: none !important;\n",
" margin-left: 0.5em;\n",
" text-align: center;\n",
" /* unfitted */\n",
" border: var(--sklearn-color-unfitted-level-1) 1pt solid;\n",
" color: var(--sklearn-color-unfitted-level-1);\n",
"}\n",
"\n",
".sk-estimator-doc-link.fitted,\n",
"a:link.sk-estimator-doc-link.fitted,\n",
"a:visited.sk-estimator-doc-link.fitted {\n",
" /* fitted */\n",
" border: var(--sklearn-color-fitted-level-1) 1pt solid;\n",
" color: var(--sklearn-color-fitted-level-1);\n",
"}\n",
"\n",
"/* On hover */\n",
"div.sk-estimator:hover .sk-estimator-doc-link:hover,\n",
".sk-estimator-doc-link:hover,\n",
"div.sk-label-container:hover .sk-estimator-doc-link:hover,\n",
".sk-estimator-doc-link:hover {\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-3);\n",
" color: var(--sklearn-color-background);\n",
" text-decoration: none;\n",
"}\n",
"\n",
"div.sk-estimator.fitted:hover .sk-estimator-doc-link.fitted:hover,\n",
".sk-estimator-doc-link.fitted:hover,\n",
"div.sk-label-container:hover .sk-estimator-doc-link.fitted:hover,
```

```
".sk-estimator-doc-link.fitted: hover {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-3);\n",
" color: var(--sklearn-color-background);\n",
" text-decoration: none;\n",
"}\n",
"\n",
"/* Span, style for the box shown on hovering the info icon */\n",
".sk-estimator-doc-link span {\n",
" display: none;\n",
" z-index: 9999;\n",
" position: relative;\n",
" font-weight: normal;\n",
" right: .2ex;\n",
" padding: .5ex;\n",
" margin: .5ex;\n",
" width: min-content;\n",
" min-width: 20ex;\n",
" max-width: 50ex;\n",
" color: var(--sklearn-color-text);\n",
" box-shadow: 2pt 2pt 4pt #999;\n",
" /* unfitted */\n",
" background: var(--sklearn-color-unfitted-level-0);\n",
" border: .5pt solid var(--sklearn-color-unfitted-level-3);\n",
"}\n",
"\n",
".sk-estimator-doc-link.fitted span {\n",
" /* fitted */\n",
" background: var(--sklearn-color-fitted-level-0);\n",
" border: var(--sklearn-color-fitted-level-3);\n",
"}\n",
"\n",
".sk-estimator-doc-link: hover span {\n",
```

```
" display: block;\n",
"}\n",
"\n",
"/* \"?-specific style due to the `<a>` HTML tag */\n",
"\n",
"#sk-container-id-2 a.estimator_doc_link {\n",
" float: right;\n",
" font-size: 1rem;\n",
" line-height: 1em;\n",
" font-family: monospace;\n",
" background-color: var(--sklearn-color-background);\n",
" border-radius: 1rem;\n",
" height: 1rem;\n",
" width: 1rem;\n",
" text-decoration: none;\n",
" /* unfitted */\n",
" color: var(--sklearn-color-unfitted-level-1);\n",
" border: var(--sklearn-color-unfitted-level-1) 1pt solid;\n",
"}\n",
"\n",
"#sk-container-id-2 a.estimator_doc_link.fitted {\n",
" /* fitted */\n",
" border: var(--sklearn-color-fitted-level-1) 1pt solid;\n",
" color: var(--sklearn-color-fitted-level-1);\n",
"}\n",
"\n",
"/* On hover */\n",
"#sk-container-id-2 a.estimator_doc_link:hover {\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-3);\n",
" color: var(--sklearn-color-background);\n",
" text-decoration: none;\n",
"}\n",
```

```

"\n",
"#sk-container-id-2 a.estimator_doc_link.fitted:hover {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-3);\n",
"}\n",
"</style><div id=\"sk-container-id-2\" class=\"sk-top-container\"><div clas
s=\"sk-text-repr-fallback\"><pre>SVC(kernel=&#x27;linear&#x27;)</pre><b>In a
Jupyter environment, please rerun this cell to show the HTML representation or tr
ust the notebook. <br />On GitHub, the HTML representation is unable to render,
please try loading this page with nbviewer.org.</b></div><div class=\"sk-contai
ner\" hidden><div class=\"sk-item\"><div class=\"sk-estimator fitted sk-toggleabl
e\"><input class=\"sk-toggleable__control sk-hidden--visually\" id=\"sk-estimator
-id-2\" type=\"checkbox\" checked><label for=\"sk-estimator-id-2\" class=\"sk-to
ggleable__label fitted sk-toggleable__label-arrow\"><div><div>SVC</div></div>
<div><a class=\"sk-estimator-doc-link fitted\" rel=\"noreferrer\" target=\"_blank\"
\" href=\"https://scikit-learn.org/1.6/modules/generated/sklearn.svm.SVC.html\">
?<span>Documentation for SVC</span></a><span class=\"sk-estimator-doc-lin
k fitted\">i<span>Fitted</span></span></div></label><div class=\"sk-toggleabl
e__content fitted\"><pre>SVC(kernel=&#x27;linear&#x27;)</pre></div> </div>
</div></div></div>
]
},
"metadata": {},
"execution_count": 23
}
]
},
{
"cell_type": "markdown",
"metadata": {
"id": "UV4-CAfquiyP"
},
"source": [

```

```
"Model Evaluation"
]
},
{
  "cell_type": "markdown",
  "metadata": {
    "id": "yhAjGPJWunXa"
  },
  "source": [
    "Accuracy Score"
  ]
},
{
  "cell_type": "code",
  "metadata": {
    "id": "fJLEPQK7ueXp"
  },
  "source": [
    "# accuracy score on the training data\n",
    "X_train_prediction = classifier.predict(X_train)\n",
    "training_data_accuracy = accuracy_score(X_train_prediction, Y_train)"
  ],
  "execution_count": 24,
  "outputs": []
},
{
  "cell_type": "code",
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/"
    },
    "id": "mmJ22qhVvNwj",
    "outputId": "e340dbf5-616b-4ff4-c012-f469d6b2b5e6"
  }
}
```

```
},
"source": [
  "print('Accuracy score of the training data : ', training_data_accuracy)"
],
"execution_count": 25,
"outputs": [
  {
    "output_type": "stream",
    "name": "stdout",
    "text": [
      "Accuracy score of the training data : 0.7866449511400652\n"
    ]
  }
],
},
{
  "cell_type": "code",
  "metadata": {
    "id": "G2CICFMEvcCI"
  },
  "source": [
    "# accuracy score on the test data\n",
    "X_test_prediction = classifier.predict(X_test)\n",
    "test_data_accuracy = accuracy_score(X_test_prediction, Y_test)"
  ],
  "execution_count": 26,
  "outputs": []
},
{
  "cell_type": "code",
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/"
    }
  }
}
```

```
},
  "id": "i2GcW_t_vz7C",
  "outputId": "f30e7339-f165-4535-fafd-47b4ed88f212"
},
  "source": [
    "print('Accuracy score of the test data : ', test_data_accuracy)"
  ],
  "execution_count": 27,
  "outputs": [
    {
      "output_type": "stream",
      "name": "stdout",
      "text": [
        "Accuracy score of the test data : 0.7727272727272727\n"
      ]
    }
  ],
},
{
  "cell_type": "markdown",
  "metadata": {
    "id": "gq8ZX1xpWPF5"
  },
  "source": [
    "Making a Predictive System"
  ]
},
{
  "cell_type": "code",
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/"
    }
  },
```

```

    "id": "U-ULRe4yv5tH",
    "outputId": "74d5aac6-59bf-4645-c753-4b60b14cfb71"
  },
  "source": [
    "input_data = (5,166,72,19,175,25.8,0.587,51)\n",
    "\n",
    "# changing the input_data to numpy array\n",
    "input_data_as_numpy_array = np.asarray(input_data)\n",
    "\n",
    "# reshape the array as we are predicting for one instance\n",
    "input_data_reshaped = input_data_as_numpy_array.reshape(1,-1)\n",
    "\n",
    "# standardize the input data\n",
    "std_data = scaler.transform(input_data_reshaped)\n",
    "print(std_data)\n",
    "\n",
    "prediction = classifier.predict(std_data)\n",
    "print(prediction)\n",
    "\n",
    "if (prediction[0] == 0):\n",
    "    print('The person is not diabetic')\n",
    "else:\n",
    "    print('The person is diabetic')
  ],
  "execution_count": 28,
  "outputs": [
    {
      "output_type": "stream",
      "name": "stdout",
      "text": [
        "[[ 0.3429808  1.41167241  0.14964075 -0.09637905  0.82661621 -0.7859\n5734\n",
        " 0.34768723  1.51108316]]\n",

```

```
"[1]\n",
"The person is diabetic\n"
],
{
  "output_type": "stream",
  "name": "stderr",
  "text": [
    "/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: U
serWarning: X does not have valid feature names, but StandardScaler was fitted
with feature names\n",
    " warnings.warn(\n"
  ]
}
],
{
  "cell_type": "code",
  "metadata": {
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